

Major Front-End Migration for GRNsight, a Web Application for Visualizing Gene Regulatory and Protein-Protein Interaction Network Models

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Engineering & Computer Science Design Showcase
May 1, 2026



Loyola Marymount University
Frank R. Seaver
College of Science
and Engineering

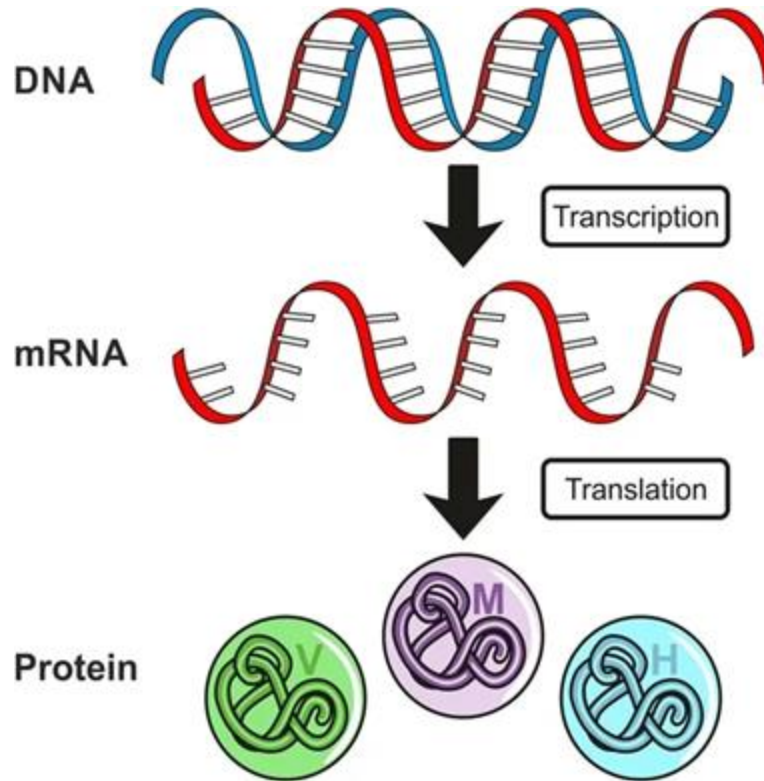
Outline

- GRNsight is an open-source tool that allows for easier visualization of Gene Regulatory Networks (GRN) and Protein-Protein Interactions (PPI).
- GRNsight is an integral part of a data analysis workflow for biologists who model GRNs.
- GRNsight's interface was originally built using older web tools, leading to repeated code and making user interface updates difficult.
- I have migrated GRNsight's user interface to React, Vite, and Grommet UI which are modern, scalable, and industry-standard frameworks.
- GRNsight React migration has resulted in user interface improvements and bug fixes that enhance the user experience for biologists.

Outline

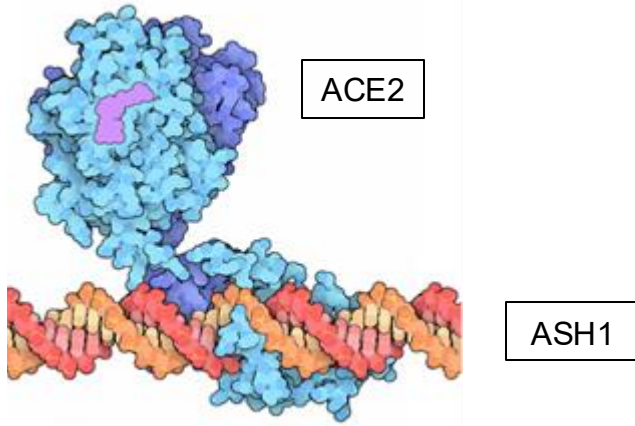
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Gene expression is the flow of information in a cell from DNA to RNA to protein

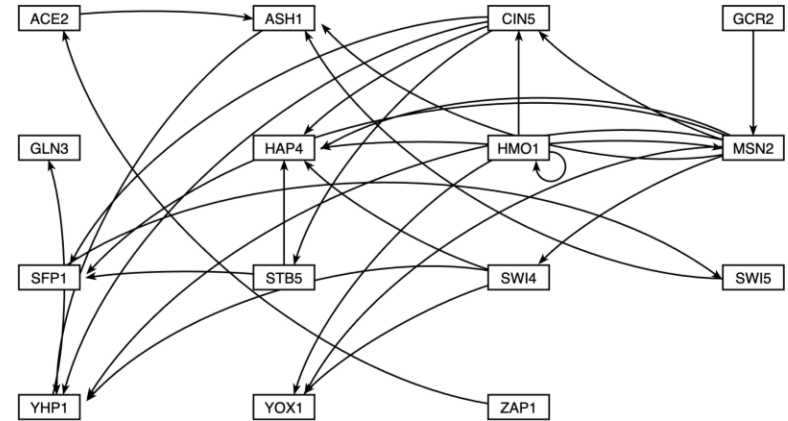
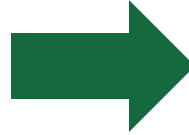


A Gene Regulatory Network (GRN) consists of genes, transcription factors and the regulatory connections between them which govern the level of expression of mRNA and protein from genes

Regulatory Transcription Factors bind to DNA and turn on (activate) or turn off (repress) gene expression



RCSB: PDB-101. <https://pdb101.rcsb.org/motm/155>



Yeast GRN that controls the cold shock response

- Node = transcription factor / gene
- Arrow edge = regulatory relationship

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



GRNsight

Web app for visualizing models of small-scale gene regulatory and protein-protein interaction networks.

A Joint Project of the
LMU Bioinformatics and
Biomathematics Groups

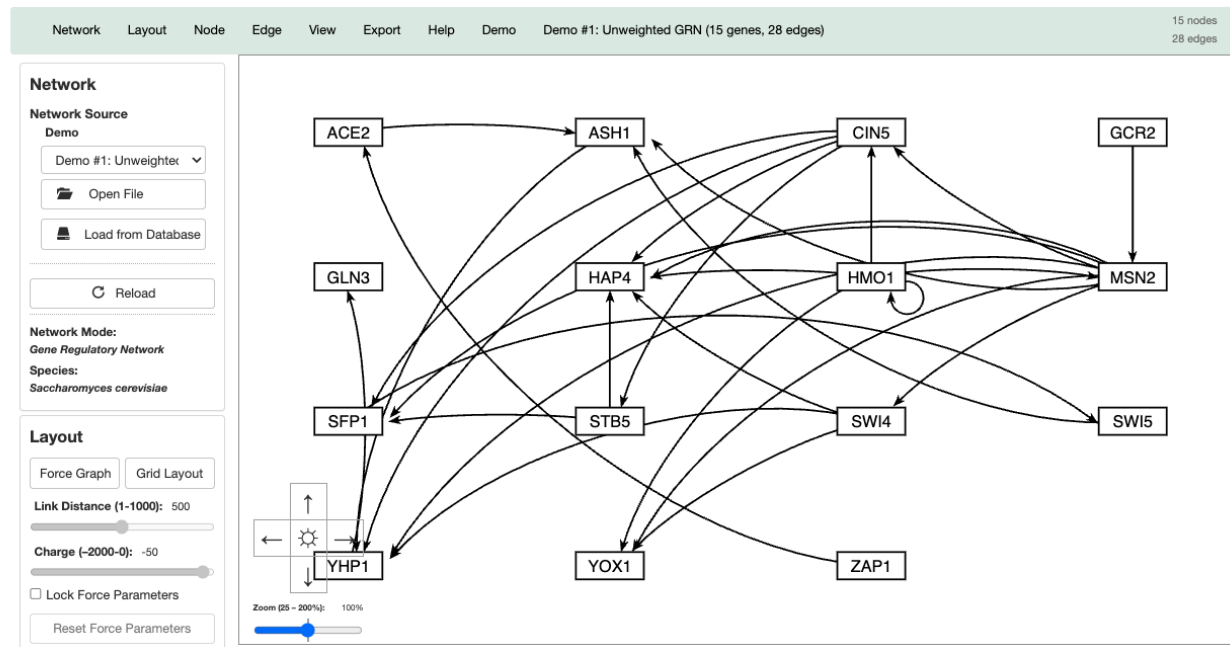
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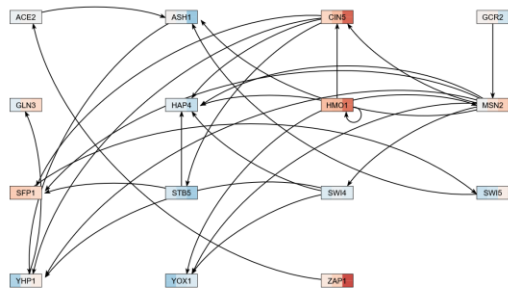
Last modified:
3/18/2026

GRNsight v7.6.0

Welcome to GRNsight. Please select a network to view.



GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Select network and expression data from GRNsight database

Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below

Network Type: Gene Regulatory

Network Source: Altancikline - SGD 2025-01-22

Select gene:

Added genes go below! Click on a gene to remove it.

ACE2 (YAL0012C)	ASH1 (YAL0012C)	CINE (YAL0012C)
GCR2 (YAL0012C)	GLN3 (YAL0012C)	HAP4 (YAL0012C)
HMC1 (YAL0012C)	MIN2 (YAL0012C)	SFP1 (YAL0012C)
STB5 (YAL0012C)	SWI4 (YAL0012C)	SWI5 (YAL0012C)
YHP1 (YAL0012C)	YCR1 (YAL0012C)	

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset: Dahlquist_2018_wt

☒ Average Replicate Values

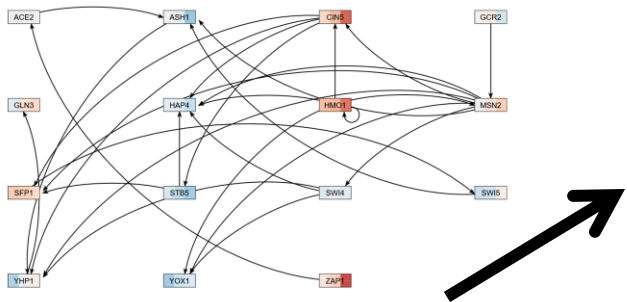
Bottom Dataset: Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value (0.01-100): 3 Set

-2.00 0 3.00

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



Export to Excel



Select network and expression data from GRNsight database



Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins

Network Type: Gene Regulatory

Network Source: Allosteric - SGD 2026-01-22

Select gene:

Add new genes by bestnet Click on a gene to remove it.

ACE2 (YGL127C)	ASH1 (YGL186W)	CIN8 (YOR058C)
GCR2 (YGL134C)	GLN3 (YGL185W)	HAF4 (YGL143W)
HMO1 (YGL117C)	MSM4 (YOR072C)	SFP1 (YGL119W)
STB5 (YOR070W)	SWI2 (YGL125C)	SWI5 (YOR146C)
YHP1 (YOR047C)	YCK1 (YGL127W)	

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

Dahlquist_2018_wt

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value (0.01-100):

3

Set

-3.50 0 3.50

Export to Excel

Select the Expression Date Source

- ☐ None
- ☐ Apowles_2012
- ☐ Bennett_2012
- ☒ Dahlgren_2016
- ☐ Khagwan_2002
- ☐ Therasme_2007

Select Workbook Sheets to Export

- ☒ Select All

Network Sheets

- ☒ network_optimized_weights
- ☒ network_weights

Expression Sheets

- ☒ d1x3_h3g1_expression
- ☒ h3g1_h3g2_expression
- ☒ d1x2_h3g1_expression
- ☒ d1x1_h3g2_expression
- ☒ w1_h3g1_expression

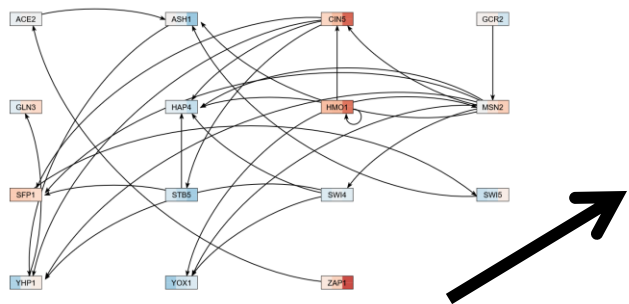
Additional Sheets

- ☒ regeneration_rates
- ☒ regeneration_parameters
- ☒ production_rates
- ☒ transmitted_h3

Export Workbook Cancel

[illegible]

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs



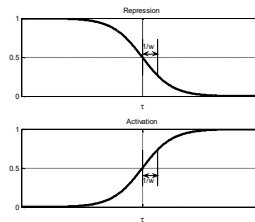
Export to Excel



Source	Target	Weight
ACE2	ASH1	1.0
ACE2	CRG	1.0
ACE2	GCR2	1.0
ACE2	GLN3	1.0
ACE2	HAP4	1.0
ACE2	HMO1	1.0
ACE2	MNR2	1.0
ACE2	SPB1	1.0
ACE2	STB5	1.0
ACE2	SWI4	1.0
ACE2	SWI5	1.0
ACE2	YHP1	1.0
ACE2	YCR1	1.0
ACE2	ZAP1	1.0

Select network and expression data from GRNsight database

Dynamical systems modeling using GRNmap



Load from Database

Generate Network

Warning: changing network type or source will clear the list of selected genes or proteins below

Network Type: Gene Regulatory

Network Source: Altitude - SGD 2025-01-22

Select gene: HAP4

Add new genes go below! Click on a gene to remove it.

ACE2 (GLN3-12)	ASH1 (HMO1-12)	CRG (HMO1-12)
GCR2 (HMO1-12)	GLN3 (HMO1-12)	HAP4 (HMO1-12)
HMO1 (HMO1-12)	MNR2 (HMO1-12)	SPB1 (HMO1-12)
STB5 (HMO1-12)	SWI4 (HMO1-12)	SWI5 (HMO1-12)
YHP1 (HMO1-12)	YCR1 (HMO1-12)	ZAP1 (HMO1-12)

Generate Network Cancel

Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

Dahlquist_2018_wt

☒ Average Replicate Values

Bottom Dataset

Same as Top Dataset

☒ Average Replicate Values

Log Fold Change Max Value
(0.01-100):

3

Set

-2.00 0 3.00

Export to Excel

Select the Expression Data Source

☐ None

☐ Agilent_2012

☐ Berman_2012

☒ Dahlquist_2018

☐ Kitzmann_2002

☐ Thomas_2007

Select Workbook Sheets to Export

☒ Select All

Network Sheets

☒ network

☐ network_optimized_weights

Expression Sheets

☒ dhlq2_log2_expression

☒ dhlq3_log2_expression

☒ dhlq4_log2_expression

☒ dhlq5_log2_expression

☒ wt_log2_expression

Additional Sheets

☒ degradation_rates

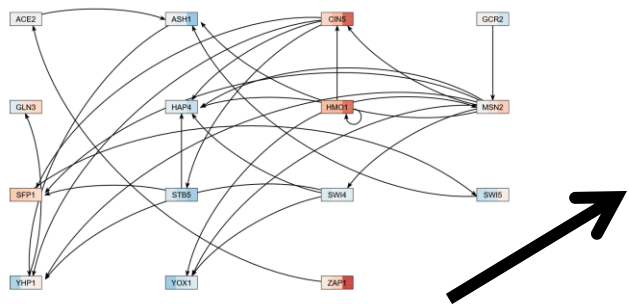
☒ optimization_parameters

☒ production_rates

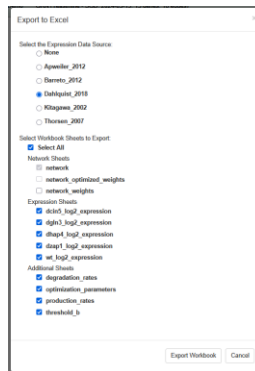
☒ threshold_b

Export Workbook Cancel

GRNsight is an integral part of a data analysis workflow for biologists who model GRNs

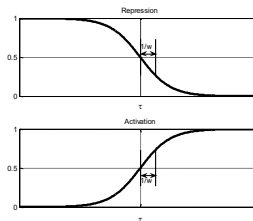


Export to Excel

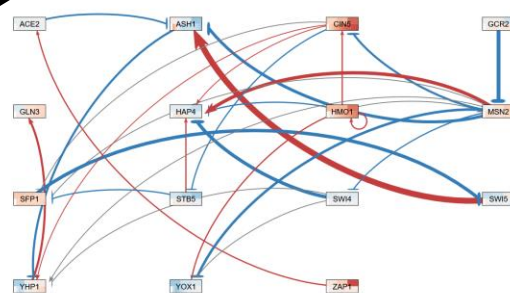


Gene	ACE2	ASH1	CRG	GCR2	GLN3	HAP4	HMO1	MSN2	SFP1	STB5	SWI4	SWI5	YHP1	YOX1	ZAP1
production rates	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
degradation rates	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
regulatory targets	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Dynamical systems modeling using GRNmap



Visualization of modeling results using GRNsight



Select network and expression data from GRNsight database



Node

☒ Enable Node Coloring

Select from user-uploaded expression data, or use data from our Expression Database

Top Dataset

Dahlquist_2018_wt

☒ Average Replicate Values

Bottom Dataset

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☒ Average Replicate Values

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3

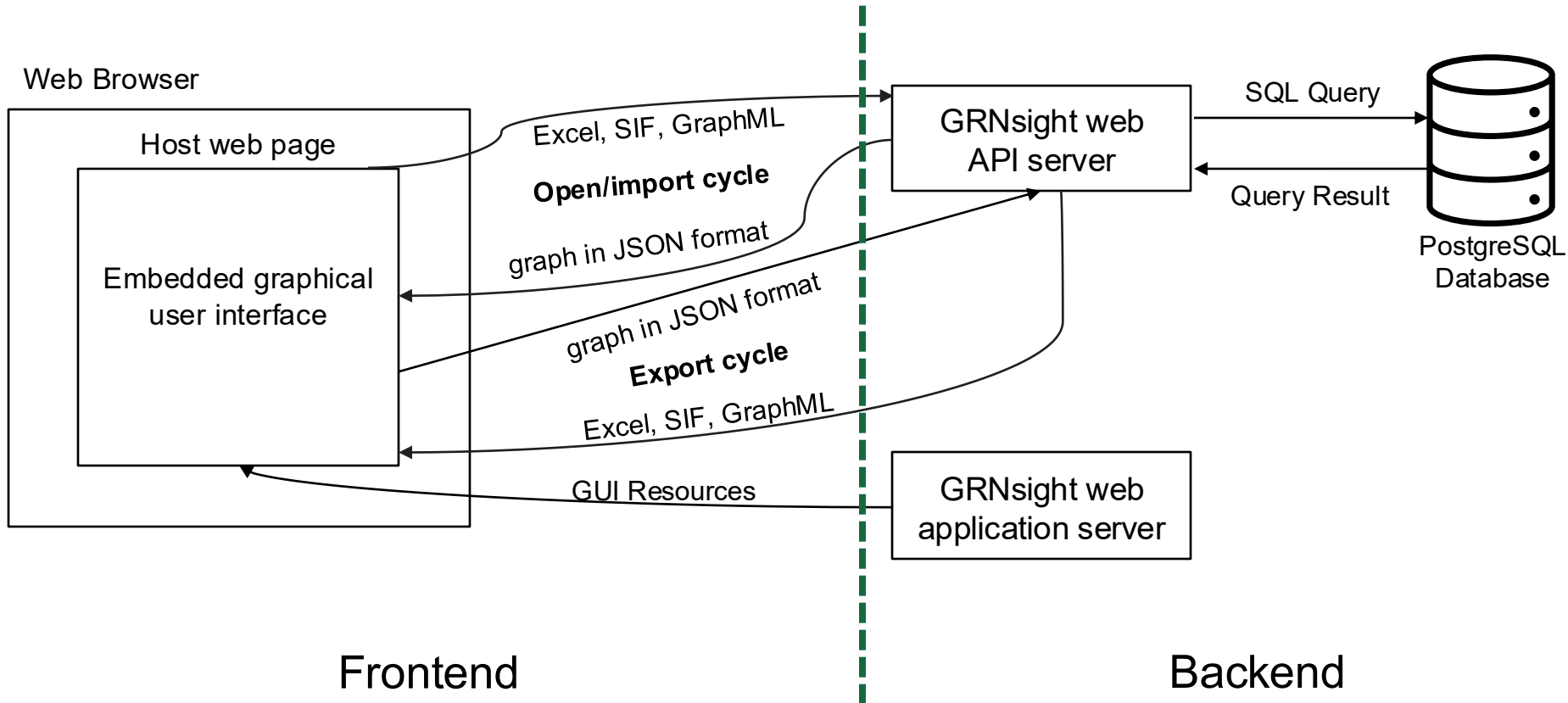
Set

-2.00 0 2.00

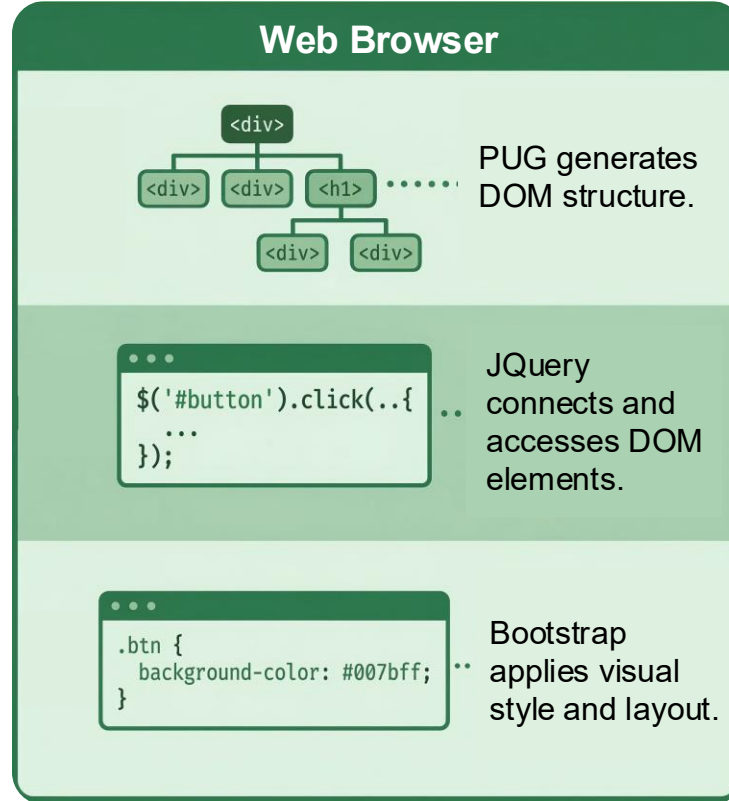
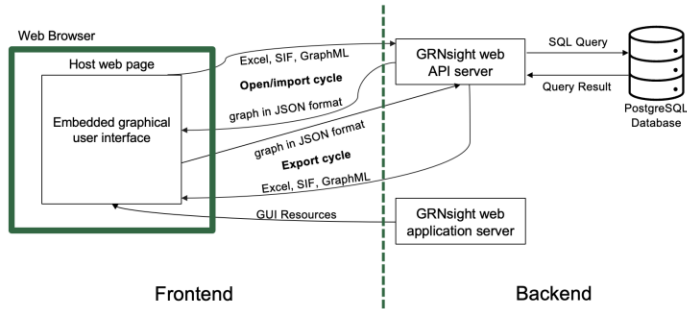
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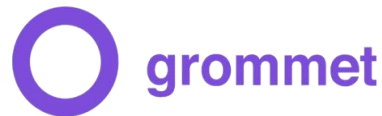
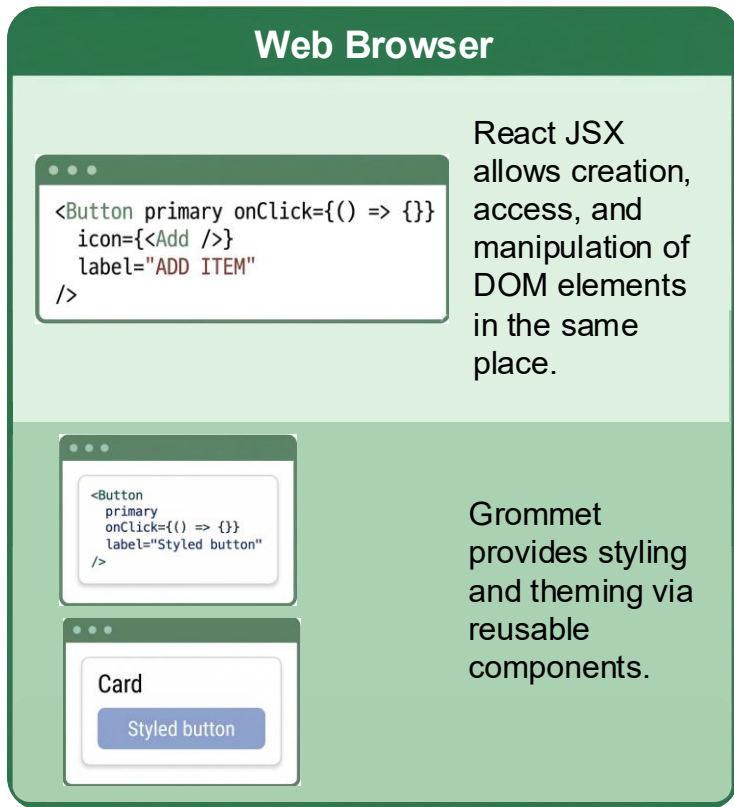
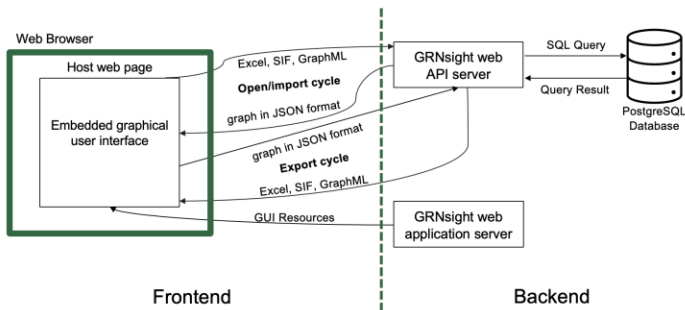
GRNsight serves the browser UI from the web application server while the API server handles import/export and all database access.



GRNsight Classic built on jQuery/Bootstrap/Pug framework for frontend resulting in code redundancy and complex UI updates



React/Grommet framework for frontend improves state management, code maintainability, and user interaction



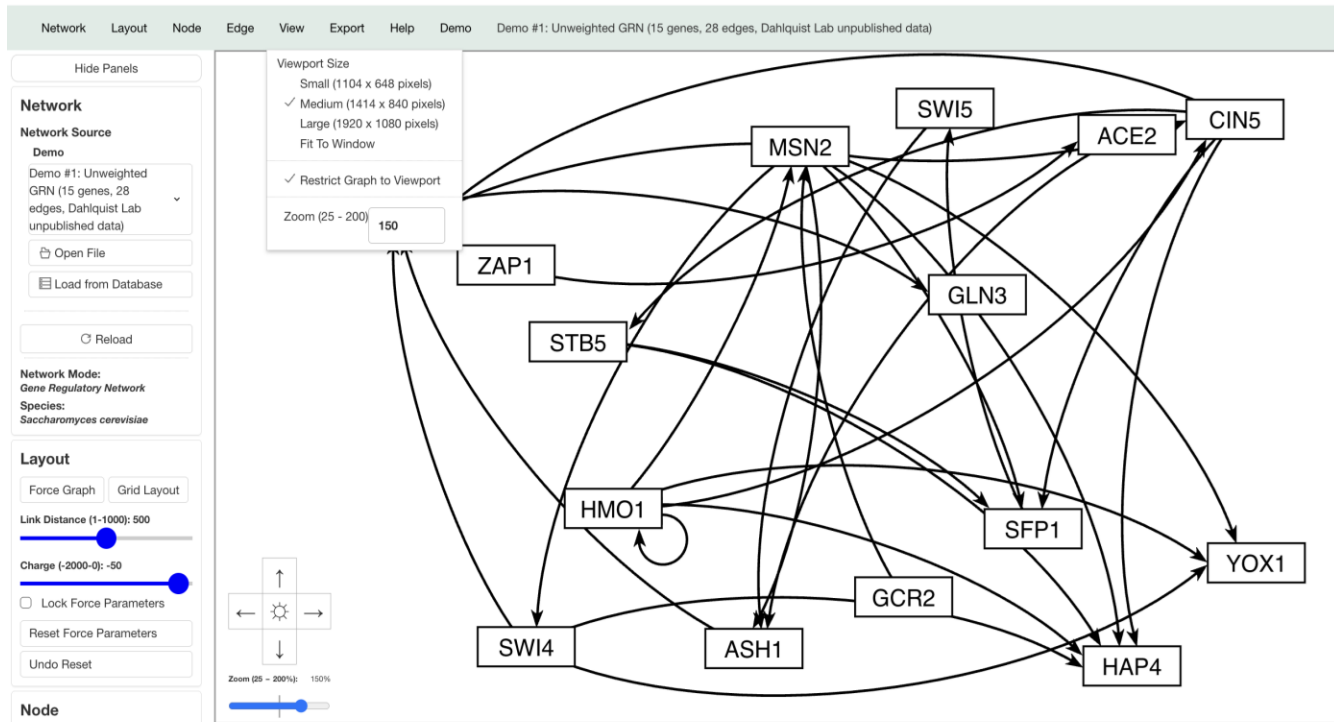
React migration resulted in tech stack improvements to developer experience and shorter build time

Feature	Classic	React / New Updates
Frontend	 Bootstrap  pug  jQuery 	 grommet  React 
Backend	 node  PostgreSQL express  JS	 node  PostgreSQL express  JS
Build Tool	 webpack	 Vite
Testing	 MOCHA  chai	 Vitest
State Management	Manual tracking of user interaction and input, and interface updates are tedious.	Automatic re-rendering of app based on user interaction and input, simplifying code.
Bug Fixes	Repetitive and complicated code makes bugs more likely and harder to find.	Modular code allows for easier discovery of bugs and fewer bugs created.

GRNsight React went live on October 15, 2025, and can now display graphs

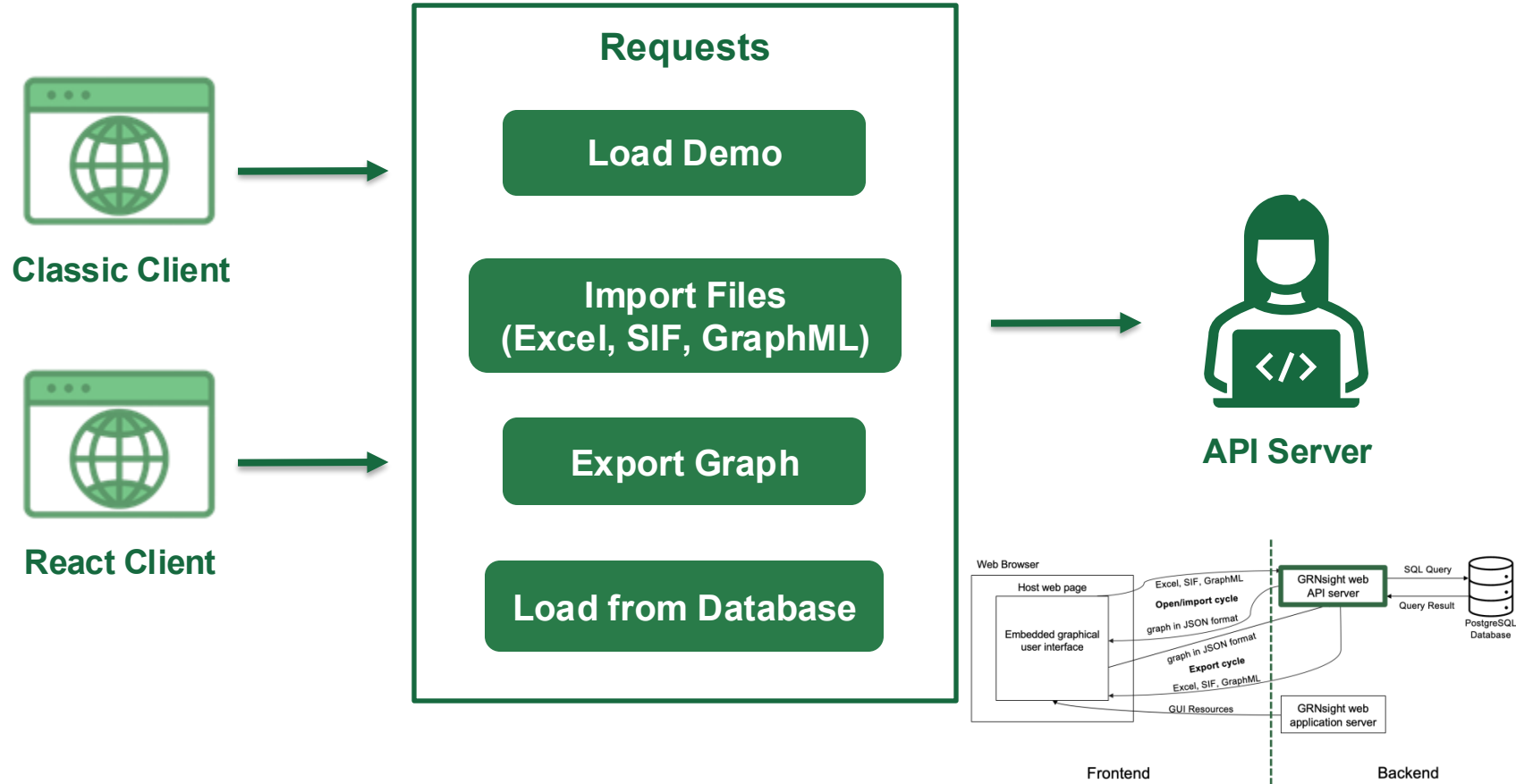
GRNsight React: v0.0.7 (04/09/2026)

Disclaimer: This version of GRNsight is currently under development and is unstable. For the most stable version, please go to the GRNsight home page.

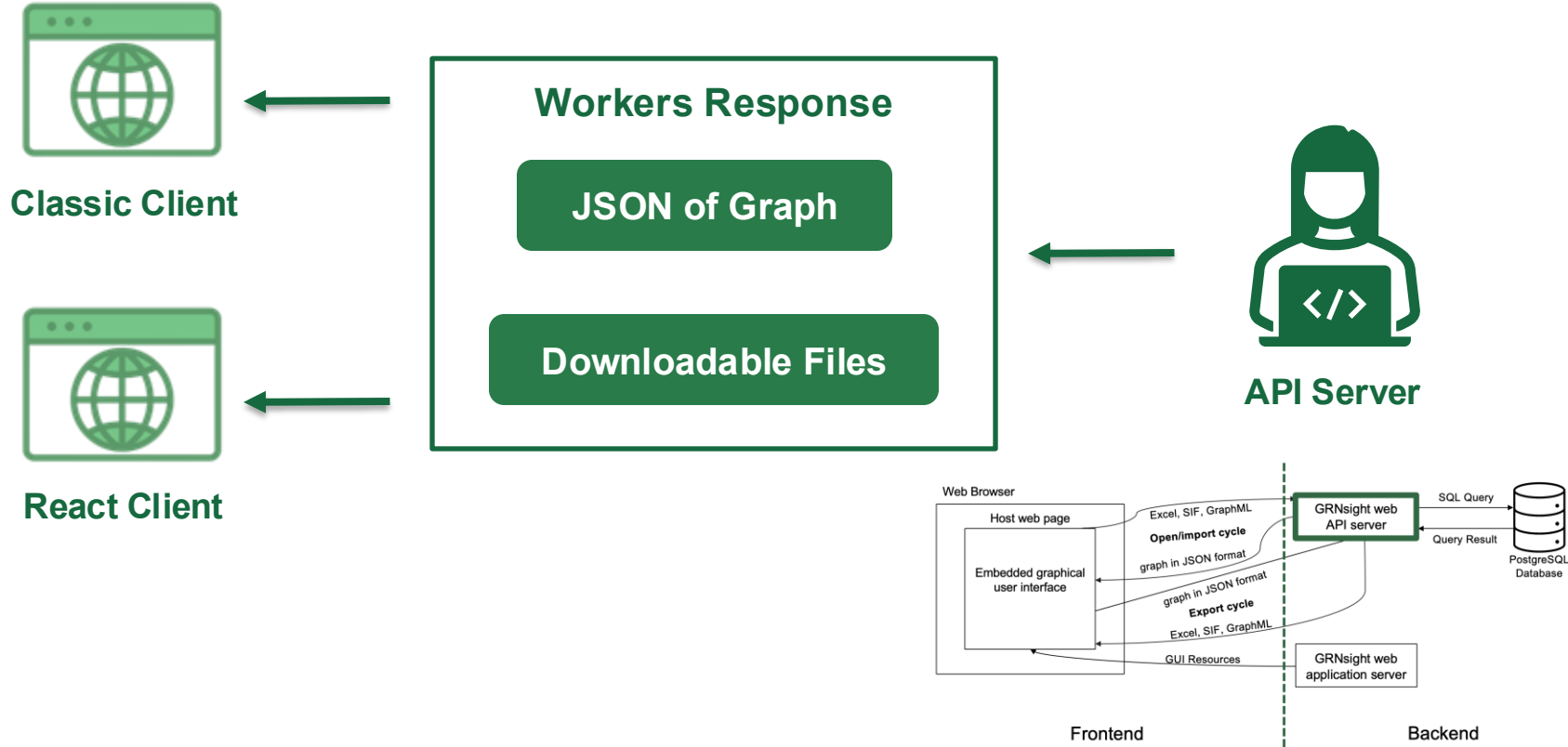


- <https://dondi.github.io/GRNsight/react-thesis-4081/>
- Viewport size, panning, and zoom are available graph features

Clients send different requests to the workers in the back



Workers process the requests and return proper results to clients



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React streamlined the fast implementation of collapsible side panels

GRNsight v7.4.1

Welcome to GRNsight. Please select a network to view.

NetworkLayoutNodeEdgeViewExportHelpDemo

Network

Network Source

Demo

Demo #1: Unweighted C ▾

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

Link Distance (1-1000): 500

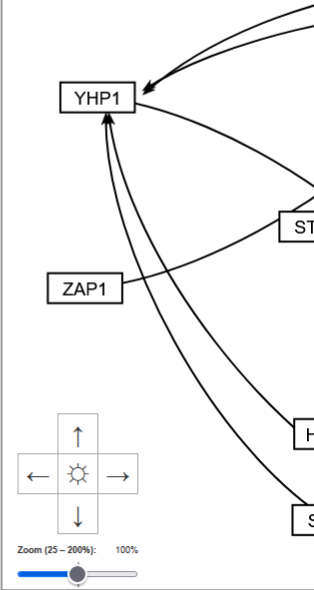
Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Undo Reset

Node



GRNsight React: v0.0.7 (04/09/2026)

Disclaimer: This version of GRNsight is currently under deve

NetworkLayoutNodeEdgeViewE

Hide Panels

Network

Network Source

Demo

Demo #1: Unweighted GRN (15 genes, 28 edges, Dahlquist Lab unpublished data) ▾

Open File

Load from Database

Reload

Network Mode:

Gene Regulatory Network

Species:

Saccharomyces cerevisiae

Layout

Force GraphGrid Layout

Link Distance (1-1000): 500

Charge (-2000-0): -50

Lock Force Parameters

Reset Force Parameters

Undo Reset

Node

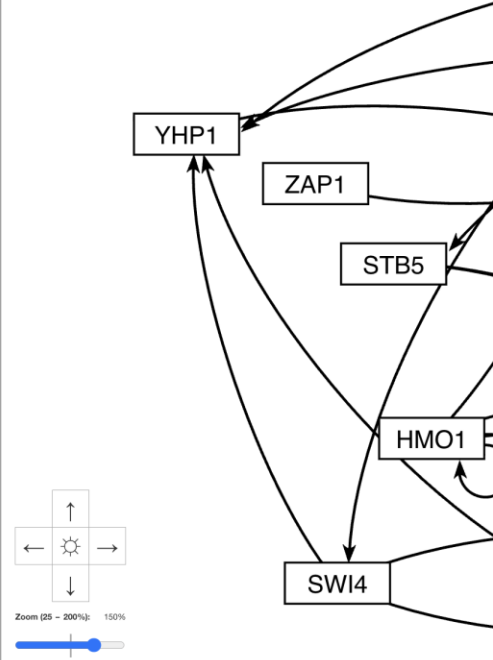
Enable Node Coloring



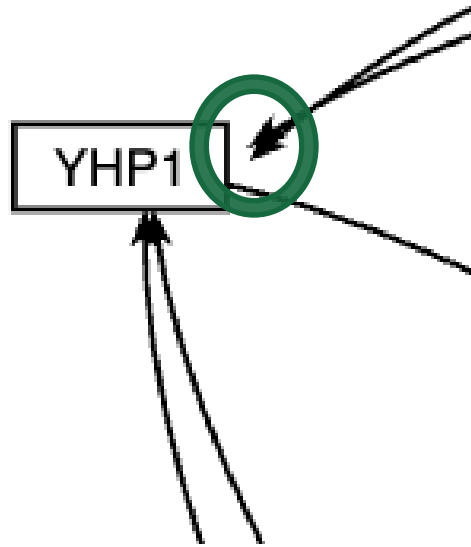
GRNsight React: v0.0.7 (04/09/2026)

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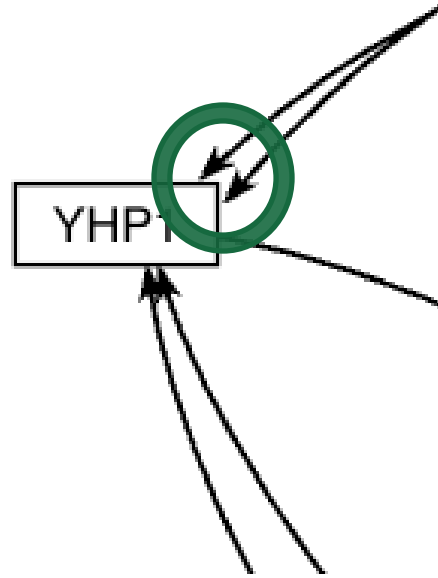
NetworkLayoutNodeEdgeViewExportHelpDemo



The high-level code review facilitated the fixing of a long-standing arrowhead bug in GRNsight Classic

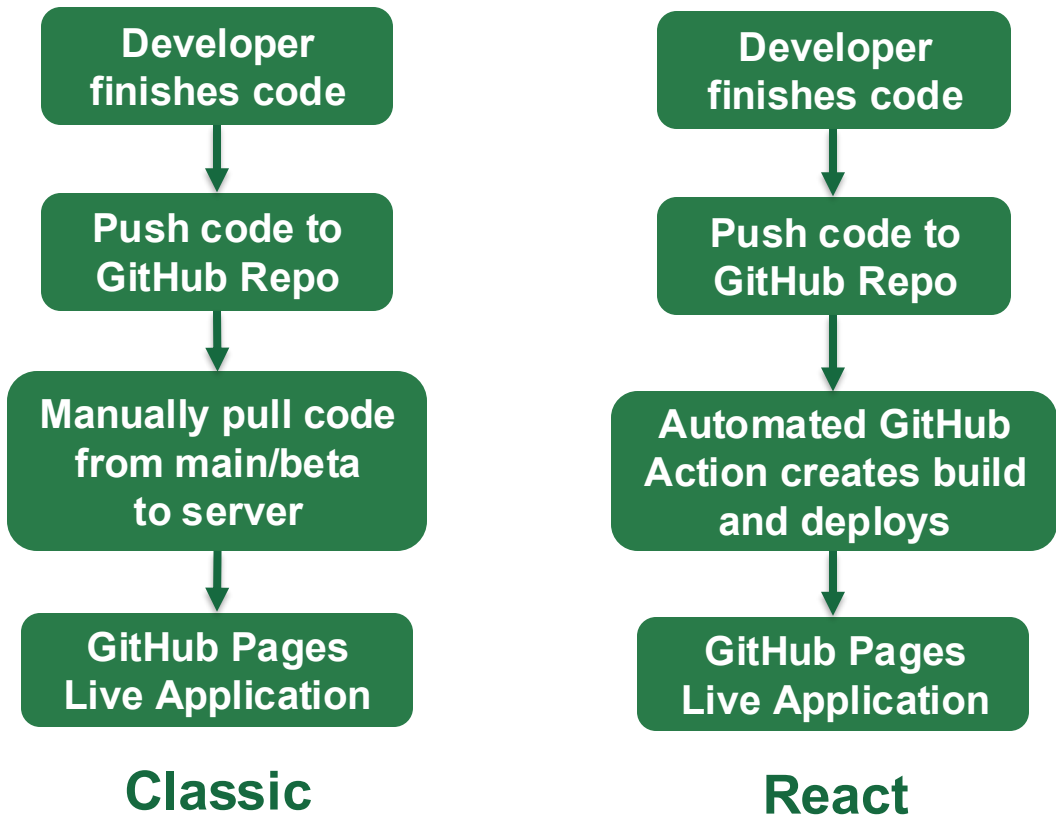


Classic Arrowheads with space between the perimeter



React Arrowheads touching the perimeter of the node

Automated deployment of React app makes release process simpler and less manual



Future Work

Migration	Future Work
 Node Coloring  Open File Tests	 Integrate React app to main GRNsight application



GRNsight turned 12 years old this year!

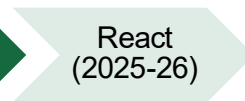
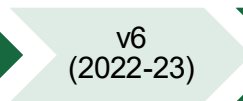
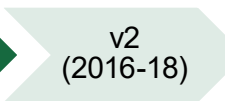


- Viewport and bounding box separated
- Zooming and scrolling
- Edge weight normalization factor

- Model-View-Controller paradigm introduced

- Backend GRN database from YeastMine

- Port to React
- Load database from Allancemine



- GRNsight server live on **March 20, 2014**
- Loads unweighted and weighted GRNs using D3's force graph layout
- Imports Excel, SIF, GraphML formats
- Demo files provided

- Import an entire GRNmap-compatible Excel workbook
- Node coloring
- Expression database
- Gene pages
- Grid layout

- Export to GRNmap-compatible Excel workbook
- Refactor expression database

- Displays PPIs
- Backend PPI database from YeastMine
- UI and export improvements
- GitHub actions

Summary

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Acknowledgements

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Kam D. Dahlquist

Alex J. Miller

John David N. Dionisio

LMU Department of Biology

HHMI STEM Research Learning Community



<http://dondi.github.io/GRNsight/>

<https://dondi.github.io/GRNsight/react-thesis-4081/>



<https://github.com/dondi/GRNsight>

References

- AZoLifeSciences. “A Guide to Understanding Gene Expression.” *AZoLifeSciences*, 27 Jan. 2021, www.azolifesciences.com/article/A-Guide-to-Understanding-Gene-Expression.aspx.
- “PDB101: Molecule of the Month: Vitamin D Receptor.” *RCSB: PDB-101*, pdb101.rcsb.org/motm/155.
- “PDB101: Molecule of the Month: Cyclin and Cyclin-dependent Kinase.” *RCSB: PDB-101*, pdb101.rcsb.org/motm/236.